United States International University-Africa

Education to take you places

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SCHOOL OF SCIENCE AND TECHNOLOGY

MIDTERM PROJECT -APT 3065

CAMPUS NAVIGATION

DECLARATION

I hereby declare that this project proposal, entitled *"Design and Development of a Campus Navigation and Mapping Application for United States International University – Africa (USIU-A),"* is my original work and has not been submitted to any other university or institution of higher learning for academic credit. Any work or contributions of other researchers, authors, or practitioners have been duly acknowledged through proper citations and references in accordance with APA guidelines.

This proposal has been prepared in partial fulfillment of the requirements for the award of the degree in Applied Computer Technology, with a concentration in Cybersecurity, at the United States International University – Africa.

I take full responsibility for the content of this document and affirm that it reflects my personal research, ideas, and efforts in conceptualizing and developing the proposed project.

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ABSTRACT

Navigating large university campuses can be a significant challenge for students, staff, and visitors, particularly in identifying the locations of lecture halls, offices, and essential service points. This project proposes the design and development of a Campus Navigation and Mapping Application for United States International University – Africa (USIU-A). The primary objective of the system is to provide a user-friendly mobile solution that enables accurate indoor and outdoor navigation within the campus while offering detailed information about key facilities.

The proposed system will integrate Firebase Firestore for backend data management and the Google Maps API for interactive mapping and routing functionalities. Users will be able to search for buildings, retrieve relevant details, and receive guided routes from their current location to the desired destination. To enhance usability, the application will incorporate a clean and intuitive interface designed in Android Studio, while also ensuring performance, scalability, and data security.

A phased methodology will be adopted in the development of this system, including research, requirements gathering, design, implementation, testing, evaluation, deployment, and documentation. The expected outcome is a reliable and efficient navigation tool that improves accessibility, reduces time wastage, and enhances the overall campus experience for students, staff, and visitors.

This project not only addresses a practical navigation problem but also contributes to the broader vision of building smart campuses by leveraging modern mobile technologies and cloud-based services

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